Additional examples for Chapter 2: 3 Problems

1. Fund M plans to raise 800 million won from new investors to invest in K Oil and H Steel stocks. The expected annual return is 30,000 won per share for K Oil and 50,000 won per share for H Steel. Currently, the stock price per share is 250,000 won for K Oil and 500,000 won for H Steel. Fund M assesses the risk index of K Oil at 0.5 per share and the risk index of H Steel at 0.25 per share and plans to maintain the total risk index within 700. Additionally, as a matter of policy, the fund plans not to invest more than 1,000 shares in K Oil. Formulate this problem as a linear optimization model.

Solution to Example 1

Decision variables and objective

Let x be the number of K Oil shares and y the number of H Steel shares purchased. Interpret the fund's objective as maximizing expected annual return, subject to its budget and risk limits.

$$Maximize\ Z\ =\ 30000x\ +\ 50000y\ (unit{:}won)$$

Subject to

$$250000x\ +\ 500000y\ \leq\ 800000000\ (unit{:}won)$$

$$0.5x\ +\ 0.25y\ \leq\ 700$$

$$x\ \leq\ 1000$$

$$x\ \geq\ 0,\ \ y\ \geq\ 0$$

If all funds must be invested, replace the budget inequality with equality. Both formulations give the same optimal portfolio here.

2. A chili pepper wholesaler operates three warehouses nationwide and supplies five cities. The transportation cost per unit (10,000 KRW/ton) from each warehouse to each city, as well as the supply and demand quantities for each warehouse and city, are given in the following table. Formulate a linear optimization model to minimize the total transportation cost.

(Unit: 10,000 KRW/ton)

City / Warehouse	A	B	C	D	E	Stock (ton)
1	7	5	12	11	9	500
2	13	12	6	3	8	300
3	7	6	5	4	14	350
Demand (ton)	150	200	100	300	400	

Solution to Example 2

Decision variables and objective

Let x_{ij} be tons shipped from Warehouse i to City j, where i = 1, 2, 3 and j = A, B, C, D, E.
Minimize Z, measured in units of 10,000 won:

$$minimize\ Z = 7x_{1A} + 5x_{1B} + 12x_{1C} + 11x_{1D} + 9x_{1E}$$
$$+ 13x_{2A} + 12x_{2B} + 6x_{2C} + 3x_{2D} + 8x_{2E}$$
$$+ 7x_{3A} + 6x_{3B} + 5x_{3C} + 4x_{3D} + 14x_{3E}$$

Subject to

$$x_{1A} + x_{1B} + x_{1C} + x_{1D} + x_{1E} \leq 500$$
$$x_{2A} + x_{2B} + x_{2C} + x_{2D} + x_{2E} \leq 300$$
$$x_{3A} + x_{3B} + x_{3C} + x_{3D} + x_{3E} \leq 350$$
$$x_{1A} + x_{2A} + x_{3A} = 150$$
$$x_{1B} + x_{2B} + x_{3B} = 200$$
$$x_{1C} + x_{2C} + x_{3C} = 100$$
$$x_{1D} + x_{2D} + x_{3D} = 300$$
$$x_{1E} + x_{2E} + x_{3E} = 400$$
$$x_{i,j} \geq 0 \ \ for\ all\ i,j$$

3. An investment firm is investing 10 billion won across two stocks: Petroleum Corporation and Steel Corporation. Based on data analysis for each company's stock, they have created the following table.

Stock	Price per share (KRW/share)	Expected Annual Return per share (KRW/share)	Risk index per share (1/share)
Petroleum Corporation	25,000	6,000	0.5
Steel Corporation	50,000	10,000	0.25

The risk index per share indicates the risk level of each stock. Petroleum Corporation appears to have a higher risk level than its peers. The firm invests subject to the constraints that the portfolio's total risk index does not exceed 70,000 and that its investment in Petroleum Corporation does not exceed 100,000 shares. Develop a linear optimization model that maximizes the total annual return.

Solution to Example 3

Decision variables and linear optimization model

Let x be Petroleum Corporation shares and y be Steel Corporation shares purchased. The objective is expected annual return in won:

$$Maximize\ Z\ =\ 6000x\ +\ 10000y\ (unit{:}\ won)$$

Subject to

$$25000x\ +\ 50000y\ =\ 10000000000\ (unit{:}\ won)$$

$$0.5x\ +\ 0.25y\ \leq\ 70000$$

$$x\ \leq\ 100000$$

$$x\ \geq\ 0,\ \ y\ \geq\ 0$$

The budget equality allocates all 10 billion won. If uninvested cash is permitted, a budget inequality can be used.